

STAT 405 Project Proposal

Team Member:

- Haoyu You (Student ID: 34735811)
- Ruohan Sun (Student ID: 59673772)

Github website:

<https://github.com/HaoyuYou/Bayesian-Statistics-Project.git>

Dataset

- main:

The dataset used in this project is the Bike Sharing Dataset from the UCI Machine Learning Repository. It contains information about daily bike rental usage in a bike sharing system from 2011 to 2012.

We use the daily dataset (`day.csv`), which includes 731 observations, where each observation represents one day. The dataset records the total number of bike rentals (`cnt`) along with several explanatory variables related to time and weather conditions.

Key variables include temperature (`temp`), humidity (`hum`), wind speed (`windspeed`), season (`season`), weather situation (`weathersit`), and whether the day is a working day (`workingday`). These variables help explain patterns in bike rental demand.

```
data <- read.csv("data/day.csv")
head(data)
```

```
##   instant      dteday season yr mnth holiday weekday workingday weathersit
## 1      1 2011-01-01      1  0   1         0         6           0          2
## 2      2 2011-01-02      1  0   1         0         0           0          2
## 3      3 2011-01-03      1  0   1         0         1           1          1
## 4      4 2011-01-04      1  0   1         0         2           1          1
## 5      5 2011-01-05      1  0   1         0         3           1          1
## 6      6 2011-01-06      1  0   1         0         4           1          1
##      temp      atemp      hum windspeed casual registered cnt
## 1 0.344167 0.363625 0.805833 0.1604460    331         654  985
## 2 0.363478 0.353739 0.696087 0.2485390    131         670  801
## 3 0.196364 0.189405 0.437273 0.2483090    120        1229 1349
## 4 0.200000 0.212122 0.590435 0.1602960    108        1454 1562
## 5 0.226957 0.229270 0.436957 0.1869000     82        1518 1600
## 6 0.204348 0.233209 0.518261 0.0895652     88        1518 1606
```

URL: <https://archive.ics.uci.edu/dataset/275/bike+sharing+dataset>

- backup:

Scientific Question:

Our main research question is:

How do weather conditions and seasonal factors influence daily bike rental demand?

More specifically, we aim to model how variables such as temperature, humidity, wind speed, season, and working day status affect the total number of bike rentals (**cnt**).

Bike sharing demand is influenced by both environmental conditions and human behavior patterns. Understanding these relationships can help improve demand prediction and support better planning for urban transportation systems.

Using Bayesian modelling allows us to quantify uncertainty in parameter estimates and make probabilistic interpretations of the effects of different predictors on bike rental demand.

Methodology:

For main question:

We plan to build a Bayesian regression model to explain and predict the number of bike rentals.

Because the response variable **cnt** is a count variable, we will consider a Poisson regression model:

$$cnt_i \sim \text{Poisson}(\lambda_i)$$

with

$$\log(\lambda_i) = \beta_0 + \beta_1 temp_i + \beta_2 hum_i + \beta_3 windspeed_i + \beta_4 workingday_i + \beta_5 season_i$$

We will place prior distributions on the regression coefficients:

$$\beta_j \sim \text{Normal}(0, 10)$$

The model will be implemented using the `simPPL` probabilistic programming framework used in this course.

Posterior inference will be performed using MCMC sampling, allowing us to estimate:

- posterior distributions of regression coefficients
- uncertainty in demand predictions
- the relative importance of different predictors

We will also perform posterior predictive checks to evaluate the model fit.

Contribution plan: